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CPSC 321

## Homework 3:

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1. One challenge was writing queries that needed more than a simple filter. For example, finding provinces with two cities over 500k population was not obvious at first. Using COUNT with GROUP BY solved it and gave a clean answer. Another challenge was avoiding duplicate results when joining tables. Adding DISTINCT or making sure both province and country matched fixed those issues.

2. Each has four provinces, and each province has four cities with realistic populations.

Borders were set only where they exist in real life. The tables are sized as follows:

country (4 rows, 4 columns)

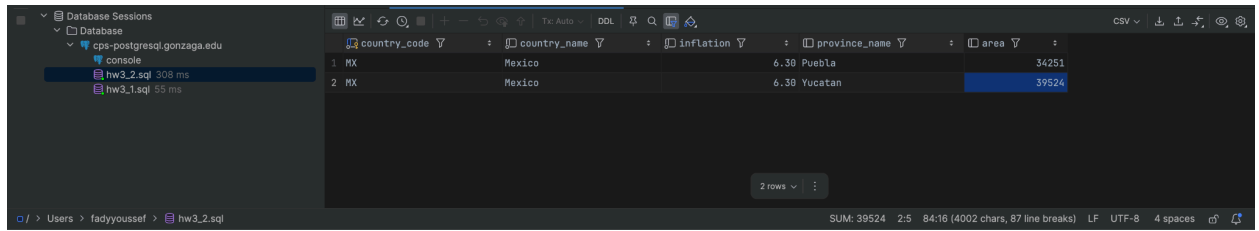
province (16 rows, 3 columns)

city (64 rows, 4 columns)

border (4 rows, 3 columns)

No extra rows were added for Part 2. The data from Part 1 already had enough variety to test queries without needing many changes.

### 3. Query: 1



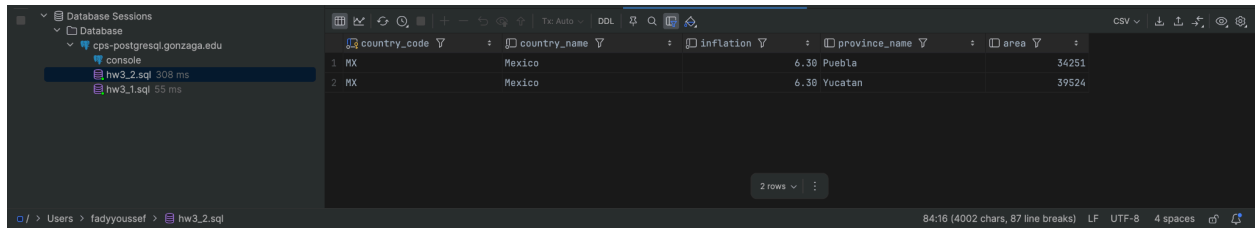
The screenshot shows a database client interface with a sidebar on the left listing sessions: 'cps-postgresql.gonzaga.edu', 'console', 'hw3.2.sql 308 ms', and 'hw3.1.sql 55 ms'. The main window displays a query result with the following columns: country\_code, country\_name, inflation, province\_name, and area. The result contains two rows.

| country_code | country_name | inflation | province_name | area  |
|--------------|--------------|-----------|---------------|-------|
| MX           | Mexico       | 6.38      | Puebla        | 34251 |
| MX           | Mexico       | 6.38      | Yucatan       | 39524 |

Example that does NOT match (small province in a low-inflation country):

INSERT INTO province VALUES ('Testshire','UK',30000);

### Query: 2



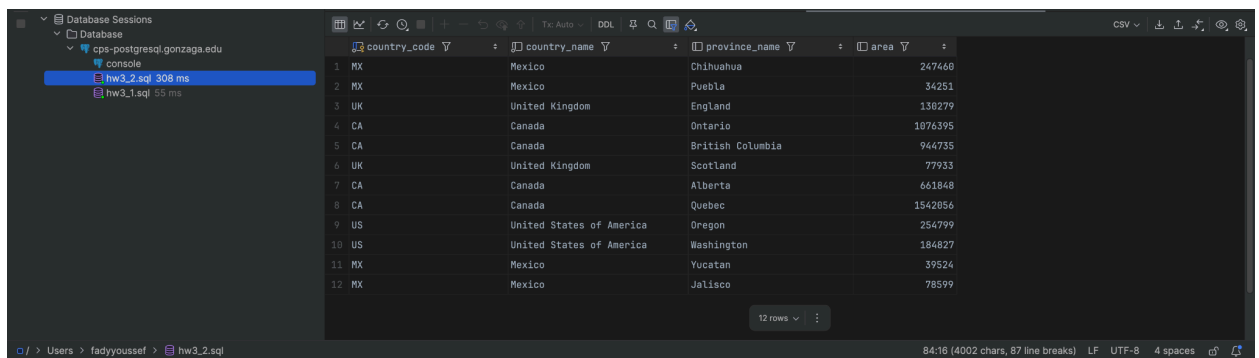
The screenshot shows the same database client interface as before, but the 'hw3.2.sql' session is selected. The query result now includes an additional row for 'Testshire' in the UK.

| country_code | country_name   | inflation | province_name | area  |
|--------------|----------------|-----------|---------------|-------|
| MX           | Mexico         | 6.38      | Puebla        | 34251 |
| MX           | Mexico         | 6.38      | Yucatan       | 39524 |
| UK           | United Kingdom |           | Testshire     | 30000 |

Example that does NOT match (same idea as Q1, JOIN version):

INSERT INTO province VALUES ('Minishire','UK',25000);

### Query: 3



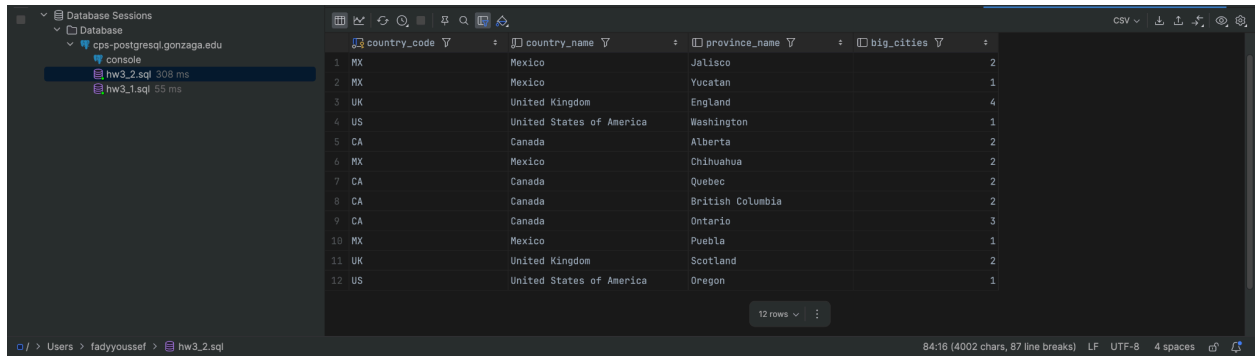
The screenshot shows the database client with the 'hw3.2.sql' session selected. The query result now contains 12 rows, including the previous data and new entries for 'Minishire' in the UK and 'Testville' in the US.

| country_code | country_name             | inflation | province_name    | area    |
|--------------|--------------------------|-----------|------------------|---------|
| MX           | Mexico                   |           | Chihuahua        | 247468  |
| MX           | Mexico                   |           | Puebla           | 34251   |
| UK           | United Kingdom           |           | England          | 138279  |
| CA           | Canada                   |           | Ontario          | 1076395 |
| CA           | Canada                   |           | British Columbia | 944735  |
| UK           | United Kingdom           |           | Scotland         | 77933   |
| CA           | Canada                   |           | Alberta          | 661848  |
| CA           | Canada                   |           | Quebec           | 1542856 |
| US           | United States of America |           | Oregon           | 254799  |
| US           | United States of America |           | Washington       | 184827  |
| MX           | Mexico                   |           | Yucatan          | 39524   |
| MX           | Mexico                   |           | Jalisco          | 78599   |
| UK           | United Kingdom           |           | Minishire        | 25000   |
| US           | United States of America |           | Testville        | 490000  |

Example that does NOT match (province with only sub-500k cities):

INSERT INTO city VALUES ('Testville','Montana','US',490000);

Query: 4

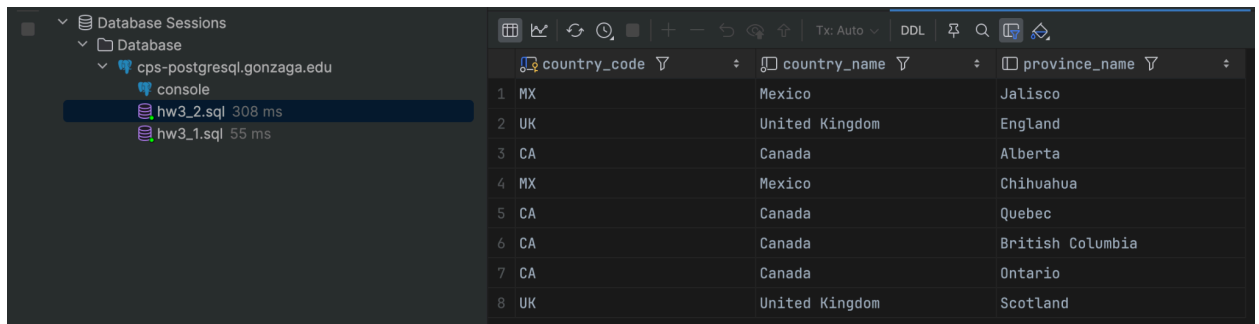


|    | country_code | country_name             | province_name    | big_cities |
|----|--------------|--------------------------|------------------|------------|
| 1  | MX           | Mexico                   | Jalisco          | 2          |
| 2  | MX           | Mexico                   | Yucatan          | 1          |
| 3  | UK           | United Kingdom           | England          | 4          |
| 4  | US           | United States of America | Washington       | 1          |
| 5  | CA           | Canada                   | Alberta          | 2          |
| 6  | MX           | Mexico                   | Chihuahua        | 2          |
| 7  | CA           | Canada                   | Quebec           | 2          |
| 8  | CA           | Canada                   | British Columbia | 2          |
| 9  | CA           | Canada                   | Ontario          | 3          |
| 10 | MX           | Mexico                   | Puebla           | 1          |
| 11 | UK           | United Kingdom           | Scotland         | 2          |
| 12 | US           | United States of America | Oregon           | 1          |

Example that does NOT match (province with zero big cities):

```
INSERT INTO city VALUES ('Smalltown','Maine','US',120000);
```

Query: 5

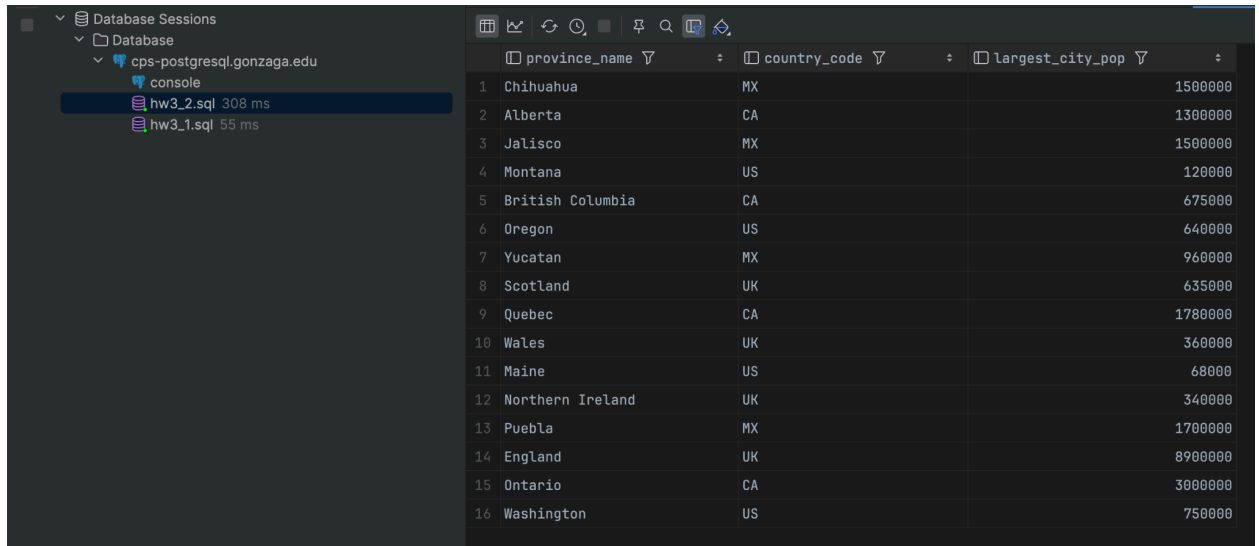


|   | country_code | country_name   | province_name    |
|---|--------------|----------------|------------------|
| 1 | MX           | Mexico         | Jalisco          |
| 2 | UK           | United Kingdom | England          |
| 3 | CA           | Canada         | Alberta          |
| 4 | MX           | Mexico         | Chihuahua        |
| 5 | CA           | Canada         | Quebec           |
| 6 | CA           | Canada         | British Columbia |
| 7 | CA           | Canada         | Ontario          |
| 8 | UK           | United Kingdom | Scotland         |

Example that does NOT match (province with exactly one big city, still < 2):

```
INSERT INTO city VALUES ('AlmostThere','Washington','US',400000);
```

## Query: 6

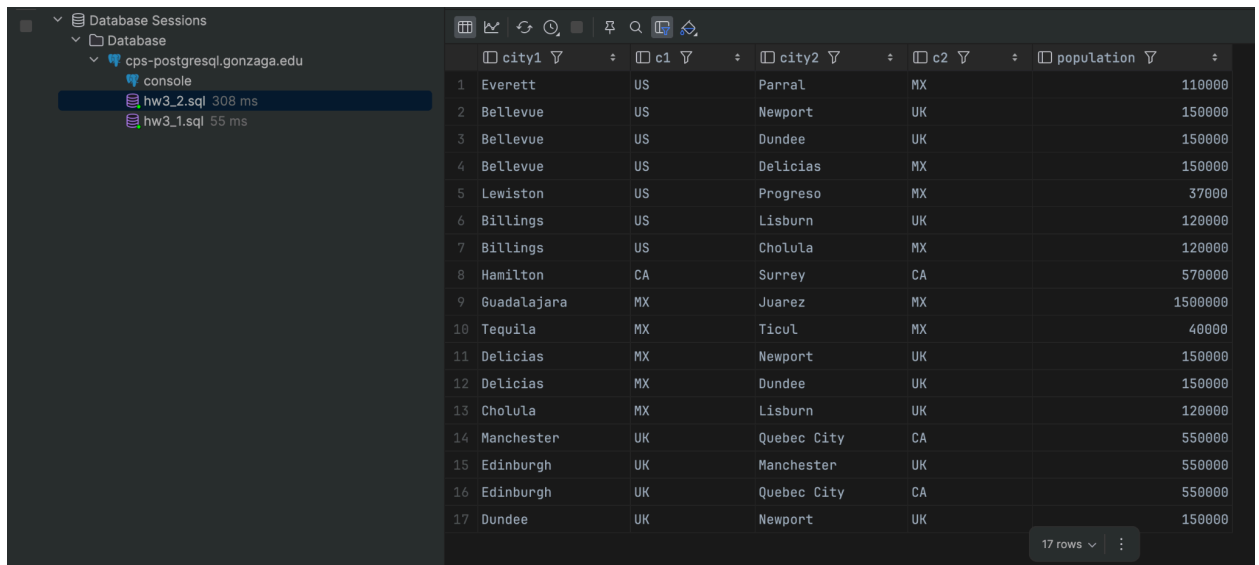


|    | province_name    | country_code | largest_city_pop |
|----|------------------|--------------|------------------|
| 1  | Chihuahua        | MX           | 1500000          |
| 2  | Alberta          | CA           | 1300000          |
| 3  | Jalisco          | MX           | 1500000          |
| 4  | Montana          | US           | 120000           |
| 5  | British Columbia | CA           | 675000           |
| 6  | Oregon           | US           | 640000           |
| 7  | Yucatan          | MX           | 960000           |
| 8  | Scotland         | UK           | 635000           |
| 9  | Quebec           | CA           | 1780000          |
| 10 | Wales            | UK           | 360000           |
| 11 | Maine            | US           | 68000            |
| 12 | Northern Ireland | UK           | 340000           |
| 13 | Puebla           | MX           | 1700000          |
| 14 | England          | UK           | 8900000          |
| 15 | Ontario          | CA           | 3000000          |
| 16 | Washington       | US           | 750000           |

Example that does NOT show up as the “max” (adds a smaller city to the province):

```
INSERT INTO city VALUES ('NotMaxCity','Quebec','CA',12345);
```

## Query: 7

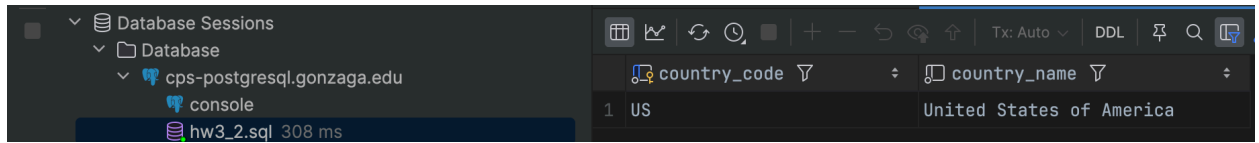


|    | city1       | c1 | city2       | c2 | population |
|----|-------------|----|-------------|----|------------|
| 1  | Everett     | US | Parral      | MX | 110000     |
| 2  | Bellevue    | US | Newport     | UK | 150000     |
| 3  | Bellevue    | US | Dundee      | UK | 150000     |
| 4  | Bellevue    | US | Delicias    | MX | 150000     |
| 5  | Lewiston    | US | Progreso    | MX | 37000      |
| 6  | Billings    | US | Lisburn     | UK | 120000     |
| 7  | Billings    | US | Cholula     | MX | 120000     |
| 8  | Hamilton    | CA | Surrey      | CA | 570000     |
| 9  | Guadalajara | MX | Juarez      | MX | 1500000    |
| 10 | Tequila     | MX | Ticul       | MX | 40000      |
| 11 | Delicias    | MX | Newport     | UK | 150000     |
| 12 | Delicias    | MX | Dundee      | UK | 150000     |
| 13 | Cholula     | MX | Lisburn     | UK | 120000     |
| 14 | Manchester  | UK | Quebec City | CA | 550000     |
| 15 | Edinburgh   | UK | Manchester  | UK | 550000     |
| 16 | Edinburgh   | UK | Quebec City | CA | 550000     |
| 17 | Dundee      | UK | Newport     | UK | 150000     |

Example that does NOT form a pair (unique population):

```
INSERT INTO city VALUES ('Uniqville','Oregon','US',1234567);
```

Query: 8



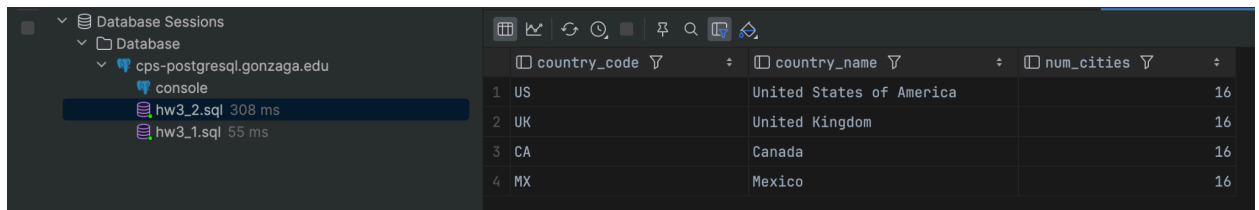
The screenshot shows a database console interface. On the left, a tree view shows 'Database Sessions' > 'Database' > 'cps-postgresql.gonzaga.edu' > 'console'. Below this, two queries are listed: 'hw3\_2.sql' (308 ms) and 'hw3\_1.sql' (55 ms). The main panel shows a query result with two columns: 'country\_code' and 'country\_name'. The result is a single row with the value 'US' for 'country\_code' and 'United States of America' for 'country\_name'.

| country_code | country_name             |
|--------------|--------------------------|
| US           | United States of America |

Example border that does NOT satisfy the high/low condition (still valid row):

INSERT INTO border VALUES ('CA','US',8891.00);

Query: 9



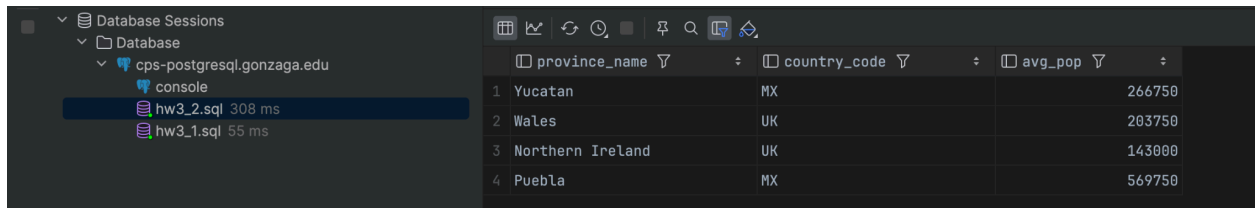
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| country_code | country_name             | num_cities |
|--------------|--------------------------|------------|
| US           | United States of America | 16         |
| UK           | United Kingdom           | 16         |
| CA           | Canada                   | 16         |
| MX           | Mexico                   | 16         |

Example country that does NOT appear (no cities for this country in dataset):

INSERT INTO country VALUES ('ES','Spain',42000,3.00);

Query: 10



The screenshot shows a database console interface. On the left, a tree view shows 'Database Sessions' > 'Database' > 'cps-postgresql.gonzaga.edu' > 'console'. Below this, two queries are listed: 'hw3\_2.sql' (308 ms) and 'hw3\_1.sql' (55 ms). The main panel shows a query result with three columns: 'province\_name', 'country\_code', and 'avg\_pop'. The result is a table with four rows.

| province_name    | country_code | avg_pop |
|------------------|--------------|---------|
| Yucatan          | MX           | 266750  |
| Wales            | UK           | 203750  |
| Northern Ireland | UK           | 143000  |
| Puebla           | MX           | 569750  |

Example province that does NOT match (area too large):

INSERT INTO province VALUES ('Bigshire','MX',50000);